Ministry of Health

Health Sector ICT Standards and Guidelines

2024

Foreword

The Ministry of Health (MOH) recognizes the value of using Information and Communication Technology (ICT) to enhance efficiency in delivering health services. ICT has the potential to impact upon every aspect of the health sector. In public health, information management and communication processes are pivotal, and are facilitated or limited by the available information and communication technology. In addition, beyond the formal health sector, the ability of impoverished communities to access services and engage with and demand a health sector that responds to their priorities and needs, is importantly influenced by wider information and communication processes, mediated by ICT.

The implementation and utilization of ICTs in healthcare in the country has been informed by the Health Sector ICT Standards and Guidelines (June 2013). However, compliance with these standards has not been optimal because implementers have worked autonomously with minimal reference to the standards. This approach leads to duplication of efforts, lack of interoperability, and inappropriate use of ICT resources, making it difficult to use ICT effectively and efficiently for healthcare service delivery.

As the Ministry of Health embarks on digitization of the health sector through the Digital Health Superhighway, it is now becoming more important to follow ICT standards to ensure effective interoperability and access to critical health information to provide citizens with high quality, consistent, accessible, and usable services. The standards set out in this document shall be applied in the health sector in the usage of ICTs by those involved in offering services or information to the public. This document provides guidance and a consistent approach across the health sector, both public and private, in establishing, acquiring, and maintaining current and future information systems and ICT infrastructure. The document utilizes the current ICT standards set by the Information and Communication Technology Authority (ICTA) to align with one government approach.

This document was developed through a participatory process involving all stakeholders in health including government ministries/agencies, development partners, and implementing partners, and utilizes the current ICT standards set by the ICTA which are based on internationally recognized best practice principles. It shall be used in accordance with the Ministry of Health Strategy and shall be reviewed and updated regularly.

It is our sincere hope that all the actors in health in Kenya will rally around these guidelines to ensure that we all steer the country towards the acceptable use of ICT.

……………………………………………

Cabinet Secretary,
Ministry of Health

Preface

The ICT Standards present a set of principles that the health sector plans to use to bring best practices in the application of ICT in service delivery. With these standards, the health sector in Kenya envisions efficient, accessible, equitable, secure and consumer friendly healthcare services enabled by ICT. The standards have been adopted from the ICT standards set by ICTA and will be updated when they are revised by ICTA. The standards address the following broad areas; Infrastructure, Systems and Applications, IT Security, Electronic Records Management, IT Governance, and ICT Human Capacity.

ICT governance presents the system by which ICT use shall be directed and controlled. It seeks to align ICT programs to enable the realization of health service delivery, exploit ICT opportunities and institute responsible use of ICT resources and appropriate management of ICT-related risks. The ICT infrastructure guidelines cover telecommunication, computing equipment, and user devices. The document outlines guidelines for achieving consistency in ICT equipment management including acquisition, management, and disposal.

Regarding software, the guidelines aim at assuring quality, internal usability, and adherence to the agreed application specific requirements. Such best practices shall significantly contribute to the successful design, acquisition, deployment, and utilization of information systems. In addition, the standards shall ensure conformity to WHO Health Informatics Standards. With Information systems being misused by the entrusted users, the standards and guidelines lay down the guidelines of proper and professional utilization of these systems and define the consequences of misuse of the same.

Information systems security aims to assure confidentiality, integrity, and availability of healthcare data. Information Systems are inherently vulnerable to attack, both physical and virtual in nature. These systems also face other potential threats, such as theft, natural and man-made calamities. Violation of privacy and unauthorized disclosure of patient information or medical records, system level attacks, software misconfigurations, and computer hacker attacks are all threats to health information. With the ever-increasing threats against IS, it is necessary to control access to and secure the environments in which these systems operate and are used. These standard guidelines advise on best practices controls and protocols that will secure the MoH information systems, to promote efficient service delivery.

The standard also offers guidelines on business continuity and disaster recovery plans for the health sector. Moreover, the documents proposition the need to ensure that backup and recovery procedures for network configurations, server configurations, and applications and databases are in place and working, and that audit logs are to be evaluated to support the recovery of data lost or modified, while ensuring protection of data from loss and destruction

……………………………………

Principal Secretary
Ministry of Health

Acronyms

MOH	Ministry of Health
ICT	Information and Communication Technologies
ICTA	Information and Communication Technology Authority
MICDE	Ministry of Information, Communications and The Digital Economy
NEMA	National Environmental and Management Authority
DDHIPR	Director Division of Health Informatics, Policy and Research

Table of Contents

1.0 Introduction

Article 43 (1) (a) of the Kenyan constitution guarantees that every citizen has the right— to the highest attainable standard of health, which includes the right to health care services, including reproductive health care. The National Health Policy further identifies specific tenets of health systems focus areas that need to be strengthened to enable response to this constitutional requirement. The National health strategic plan specifies ICT as a catalyst to attaining efficiency in multiple facets of the above areas.

As the Health Sector continues to deepen the devolution of health services, risks of adopting a diversified approach where ICT implementations occur without any reference to a common set of standards and guidelines have emerged. This presented challenges in integration at inter and intra-county and across the two levels of government.

Therefore, as departments responsible for Health between the two levels of Government continue to increasingly embrace ICT in service delivery, it is therefore necessary to strengthen a common approach on ICT implementation based on recognized best practices and standards.

ICT capacity in the Public Health Sector has grown as demonstrated by implementation of various HIS systems such as DHIS, MFL, EMR, eCHIS, iHRIS among other systems. ICT infrastructure has improved through the installation of Local Area Networks, National Fibre Optic Network, Data Centres and Increased cellular network coverage and satellite internet from investments by both public and private sectors. These ICT investments have led to improved service delivery and enhanced information exchange within the health sector.

The challenge that persists in delivering ICT services is ensuring consistency in implementation and harmonization of health sector system requirements. The objective of these ICT standards and guidelines is to ensure consistency in ICT initiatives and management to achieve efficiency thereby improving the delivery of healthcare.

The Ministry of Health will continuously enhance its organizational capacity by adopting modern technologies and skills development. These standards will guide both levels of government to ensure that ICT resources are optimally used to achieve efficiency in healthcare delivery.

The standards promote principles that guide implementation of robust ICT infrastructure, Information systems, support services and operational capacity.

The standards and guidelines derive the authority from:

1. Kenya Communications Act 2009
2. E-Government Strategy - 2004
3. The National ICT Policy - 2019
4. Digital Health Act - 2023
5. Data Protection Act - 2019
6. Computer Misuse and Cyber Crimes act - 2018
7. Kenya Information and Communication Act - 1998

Any other relevant legal provision and Government policies that may come into force after initial implementation of these standards and guidelines

To provide leading edge, integrated ICT solutions derived from identified strategies based on defined technologies and processes for effective and efficient health care delivery and operating excellence

To maximize MoH ability to realize its Mission – innovatively, creatively, efficiently, and effectively – and to add value to health care delivery services in order to improve health and well-being of Kenyans.

Specific objectives

1. Support the development, implementation, and maintenance of ICT Systems in MOH;
2. Enhance information security of MOH ICT systems.
3. Promote efficient and effective operations and usage of ICT systems within the MOH;
4. Encourage innovations in technology development, use of technology and general workflows within the MOH;
5. Facilitate the development of ICT skills to support ICT systems in the MOH;
6. Promote efficient communication among the MOH's staff and stakeholders;
7. Promote information sharing, transparency, and accountability within MoH and towards the public and other stakeholders.

The ICT standards shall apply to the MOH at the national and county level and its stakeholders in relation to all MOH ICT related operations. This will also cover private health providers through the various regulatory bodies including The Medical Practitioners and Dentists Board, The Pharmacy and Poisons Board, The Nursing Council, the Kenya Medical Laboratory Technologists and Technicians Board among other health regulatory bodies.

This standard shall be guided by the following key principles:

1. Mainstreaming of ICT in the Ministry
2. Integration of ICT systems
3. Adherence to best practices & policies
4. User, customer, patient satisfaction

`

The overall responsibility of implementing this standard will lie with the Principal Secretary in collaboration with the ICT Governance Committee which will be responsible for the overall strategic management of ICT resources in the Ministry. The committee will draw representation from heads of departments and the head of ICT being secretary. The committee will be responsible for oversight, enforcement and review of the standards and the initiation of ICT projects.

2.0 The ICT Standards

The last published version of the *health sector ICT standards and guidelines*[1] was published in June-2013. This was a comprehensive document that covered 9 domains to consider in ICT adoption and implementation. The areas include:

1. ICT governance
2. ICT Infrastructure
3. Virtualization, thin client and cloud computing
4. Application and systems software
5. Acceptable use of electronic communication
6. IT Security
7. Data back-up and recovery
8. Human resource development
9. Monitoring and evaluation

Subsequently, the Ministry of Information Communication and Digital Economy (MICDE), under the government enterprise architecture (GEA) framework and through ICTA established ICT standards to be applied across all levels of government (ministries, counties, and agencies). This work has identified 12 standards in 6 domains that must be considered. The third edition of the ICT standards was approved in May 2023 and became effective in July 2023. These are valid for 3 years. The defined standard and guidelines domains are:

1. Infrastructure
2. Systems and Application
3. IT Security
4. Electronic Records Management
5. ICT Governance
6. ICT Human Capacity

Following the recent call by MOH to review the existing ICT standards and guidelines, it was evident that MICDE through ICTA has conducted substantial work in developing ICT standards. In addition, the call emphasized the need to identify and adapt what presently exists in defining the revision process. Therefore, the working group focusing on these standards did a comparison between the earlier document developed by MOH and what has been done by ICTA.

[1] https://drive.google.com/file/d/1wagGSrc-cRTOdfUW7np3K0GdUgdCsFpO/view?usp=sharing

Most of the domains that were covered in the health sector ICT standards and guidelines, have been covered under different domains of the ICT standards developed by ICTA. A key component that has not been addressed is the e-waste disposal. This document has adopted practices as guided by the National Environmental Management Authority (NEMA).

The table below summarizes the 6 domains and 12 standards with links to the respective documents for reference.

2.1 ICT Standards Summary

S/No	Thematic Area	Standards	Brief Description	Live Link
		ICTA.2.1.003:2023 Network Standard	Provides compliant requirements for design, installations, and management of all categories of IT Networks to be deployed in government	https://cms.icta.go.ke//sites/default/files/2023-07/ICT%20Networks%20Standard%202023.pdf
		ICTA.2.2.002:2023 Data Center Standard	Provides compliant requirements for design, installations, and management of government data centers	https://cms.icta.go.ke//sites/default/files/2023-07/Data%20Center%20Standard-Third%20Edition%202023.pdf
		ICTA- 8.003:2023 Cloud Computing Standard	Provides compliant requirements for design, installations, and management of cloud computing infrastructures for government	https://cms.icta.go.ke//sites/default/files/2023-07/Cloud%20Standard%202023.pdf
1	Infrastructure	ICTA-2.3.002: 2023 End-User Equipment Standard	Provides the minimum	https://cms.icta.go.ke//sites/default/file

			specifications for all computing devices being deployed in government	s/2023-07/End-User%20Computing%20Devices%20Standard%202023.pdf
		ICTA. 2.001: 2021: Fiber Optic-Backbone, Metro and Last Mile Infrastructure Standard	This Standard sets out minimum requirements for the planning, design, deployment, operation, maintenance, and management of back- bone, metro, and last mile fiber optic cable.	https://cms.icta.go.ke//sites/default/files/2022-08/Fiber%20Standard.pdf
2	Systems & Applications	ICTA.6.003:2023 Systems & Applications Standard	Provides compliant requirements for design, installations and management of all government Software and applications Systems.	https://cms.icta.go.ke//sites/default/files/2023-07/Systems%20%26%20Applications%20Standard-2023.pdf
		ICTA-4.1.003:2023 Electronic Records Management System Standard	Provides compliant requirements for management of government	https://cms.icta.go.ke//sites/default/files/2023-07/ERM%20System

			electronic records and data.	s%20Standard%202023.pdf

S/No	Thematic Area	Standards	Brief Description	Live Link
3	IT Security	ICTA-3.003:2023 Information Security Standard	Provides compliant requirements for design, installations, and management of Information Technology Security in government.	Government ICT Standards
4	Electronic records management	ICTA.4.2.003:2023 Electronic records Management Standard	Provides compliant requirements for management of government electronic records and data.	https://cms.icta.go.ke//sites/default/files/2023-07/Electronic%20Records%20Management%20Standard%20-2023_0.pdf
		ICTA-4.3.003:2023: Digitization of public Records	This standard shall guide conversion and the migration of public records at National and County governments to support automation of business processes. It focuses on procedures and methodologies for changing records from one format to another and moving from one media to another.	https://cms.icta.go.ke//sites/default/files/2023-07/Digitization%20of%20Public%20Records%20standard%202023.pdf
5	ICT Governance	ICTA. 5.003.2023 ICT Governance Standard	Provides compliant requirements for IT Governance in government. This includes compliance requirements for government IT service providers and Professional Staff.	https://cms.icta.go.ke//sites/default/files/2023-07/ICT%20Governance

				%20Standard%202023.pdf
6	ICT Human Capacity	ICTA. 7.003: 2023 ICT Human Capital and Workforce Development Standard	Provides compliant requirements for development of Human Capital capacity for deployment and support for government ICT infrastructure and services.	https://cms.icta.go.ke//sites/default/files/2023-07/ICT_Human_Capital_and_workforce_development_standard-2023.pdf

2.2 Disposal ICT waste

E-waste (ICT waste) is considered hazardous waste as it contains toxic materials or can produce toxic chemicals when treated inappropriately. Utilization of ICT in health generates e-waste as devices get to the end of useful life. Disposal of ICT waste shall therefore be guided by the regulations and guidelines developed by NEMA. All users of ICT resources supporting health should adhere to the guidelines and regulations as provided by NEMA. All ICT waste must be disposed of by accredited e-waste processors. The details of e-waste disposal are explained in the Guidelines for e-Waste Management in Kenya (December 2010) and the Draft Environmental Management and Coordination (e-Waste Management) Regulations 2013. The disposal guidelines shall be updated as and when updated international standards or NEMA standards become available.

3.0 Review

This standard will be regularly reviewed and amended as required to ensure it remains relevant and effective in meeting its objectives. The responsibility for the ongoing review resides with the head of ICT in conjunction with the ICT Governance Committee. Any changes to these standards and guidelines shall be communicated to all users of the MOH ICT systems. Monitoring of the use of the standards will be done every quarter.

4.0 Contributors

The coordination and development of this document was a collaborative effort led by the MoH in collaboration with its implementing partners. The key contributors to the document were:

	Name	Organization
1.	Dr. Bernard Langat	MoH – Director - DDHIPR
2.	Dr. Ryan Nyotu	MoH - Digital Health
3.	Dr. Job Nyangena	MoH - Health Informatics
4.	Dr. Joyce Wamicwe	MoH - Policy & Research
5.	Dr. Wesley Ooga	MoH - eHealth
6.	Dr. Martha Muthami	MoH – Health Systems
7.	Jeremiah Mumo	MoH - Digital Health
8.	Diana Kamar	MoH - Digital Innovations
9.	Sophy Karanja	MoH - Health Informatics
10.	Nancy Amayo	MoH - HIS
11.	Gilbert N. Mboro	MoH - HRIOM
12.	Paul Malusi	MoH - DDHIPR
13.	Grace Kiragu	MoH Digital Health
14.	Samuel Kamau	MoH - Digital Health
15.	Susan G. Kihiu	MoH - DDHIPR
16.	Meboh Atieno	Council of Governors
17.	Dr. Tracy Njonjo	MoH – Health Commodities Supply Chain
18.	Dr. Jarred Nyakiba	MoH – DDPS
19.	George Owiso	I-TECH Kenya
20.	Stephen Wanyee	Intellisoft
21.	Margaret Ndisha	PATH
22.	Sam Wambugu	PATH
23.	Dr. Ramona Koech	Tony Blair Institute for Global Change
24.	Joan Ngéthe Cherono Kirui	Health IT
25.	Samuel Kang'a	Medtronic Labs
26.	Derick Lungáho	Medic
27.	Winfred Nzioka	Vital Strategies
28.	Charles Bin Atelu	I-TECH Kenya
29.	Dr. Karimi Muchiri	I-TECH Kenya
30.	Dr. Calvine Lwaka	EGPAF
31.	Eric Ejimba	CHISU
32.	Lawrence Kimathi	Savannah Informatics
33.	Dr. Nelly Nyaga	Intellisoft
34.	Joseph Warero	Afya Ugavi
35.	Gitahi Ng'ang'a	Living Goods
36.	Lesley Githinji	Living Goods

37.	Dr. Emma Waiyaiya	HELINA
38.	Onesmus Kamau	Living Goods
39.	Charles Brian Bett	Palladium
40.	Danson Koske	Palladium
41.	Patrick Wangai	Palladium